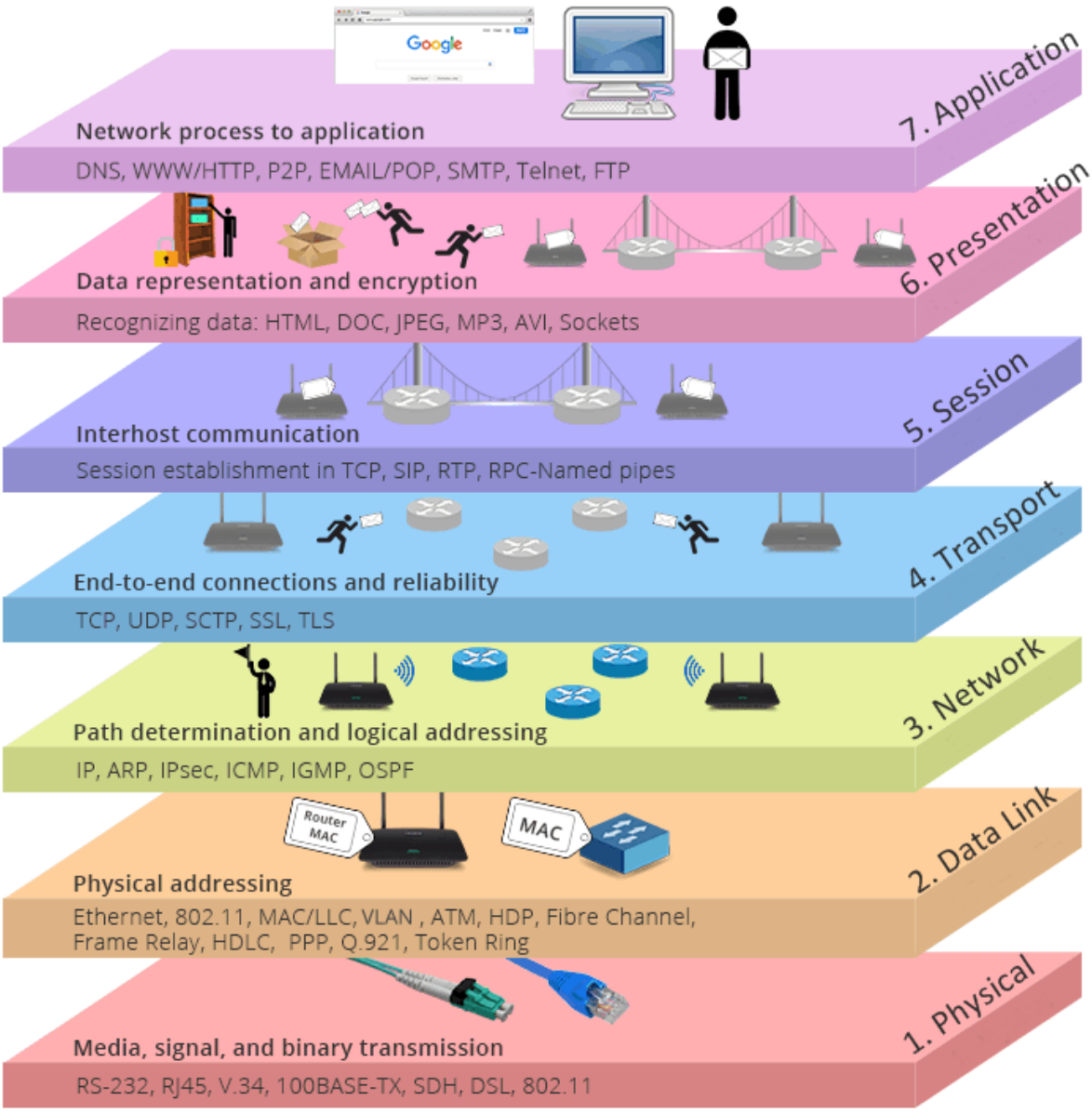
OSI Model

1. Diagram of the OSI 7-Layer Model



Data moves from Layer 7 to Layer 1 when sending and from Layer 1 to Layer 7 when receiving.

2. Match the Scenarios to OSI Layers

a) Encrypting data before sending: Presentation Layer (L6)

b) Routing packets across networks: Network Layer (L3)

c) Converting data to electrical signals: Physical Layer (L1)

d) Opening a web page in your browser: Application Layer (L7)

e) Detecting and correcting errors in frames: Data Link Layer (L2)

3. How OSI Layers Work When Sending a WhatsApp Message

When I send a message through WhatsApp, several OSI layers work together:

1. Application Layer (Layer 7): WhatsApp allows me to type and send a message.
2. Presentation Layer (Layer 6): The data is represented in a suitable format and encrypted to protect its contents. In practice, WhatsApp implements encryption within its application protocols.
3. Transport Layer (Layer 4): Transport protocols carry data between endpoints. WhatsApp commonly uses TCP-based connections for its messaging services.
4. Network Layer (Layer 3): IP addresses help route packets from my phone across networks toward the destination server.
5. Data Link and Physical Layers (Layers 2 and 1): Wi-Fi or mobile network technology transmits the data as signals over the local connection.

The receiving side processes the incoming data so the message can be displayed to the recipient.

4. Website Not Opening, but Its IP Address Responds to Ping

Possible problem: Application Layer (Layer 7), or another service-related layer.

Reasoning:

- If ping to the website's IP address succeeds, IP connectivity is working at least well enough for ICMP traffic to reach the host and return.
- The problem may be at the Application Layer (Layer 7) if the web server or website service is unavailable.
- The Transport Layer (Layer 4) may also be involved if TCP connections to ports 80 (HTTP) or 443 (HTTPS) are blocked or failing.
- DNS may be the issue if the website works by IP address but not by its domain name. DNS is an application-layer protocol.

Conclusion: Successful ping does not guarantee that a website is working. Check DNS resolution, the web server, firewall rules, and HTTP/HTTPS connectivity to identify the cause.